In fact, *almost all* the in-context learning in small attention-only models appears to come from these induction heads! This begins at the start of the phase change, and remains true through the end of training.[17]

Unfortunately, we do not have ablations for our full-scale models.[18] For the models where we do have ablations, it seems like this evidence is clearly dispositive that induction heads increase in-context learning (at least as we've chosen to evaluate it). But can we further infer that they're the primary mechanism? A couple considerations:

- In attention-only models, in-context learning must essentially be a sum of contributions from different attention heads.[19] But in models with MLPs, in-context learning could also come from interactions between MLP and attention layers. While ablating attention heads would affect such mechanisms, the relationship between the effect of the ablation on in-context learning and its true importance becomes more complicated.[20] As a result, we can't be fully confident that head ablations in MLP models give us the full picture.
- Our ablations measure the marginal effects of removing attention heads from the model. To the extent two heads do something similar and the layer norm before the logits rescales things, the importance of individual heads may be masked.

All things considered, we feel comfortable concluding from this that induction heads are the primary mechanism for in-context learning in small attention-only models, but see this evidence as only suggestive for the MLP case.

Argument 4: Despite being defined narrowly as copying random sequences, induction heads can implement surprisingly abstract types of in-context learning.

STRENGTH OF ARGUMENT FOR SUB-CLAIMS

	Small Attention-Only	Small with MLPs	Large Models
Contributes Some			Plausibility
Contributes Majority			Plausibility

All of our previous evidence (in Arguments 1-3) focused on observing or perturbing the connection between induction head formation and macroscopic in-context learning. A totally different angle is to just find examples of induction heads implementing seemingly-difficult in-context learning behaviors; this would make it plausible that induction heads account for the majority of in-context learning. This evidence applies even to the very largest models (and we study up to 12B parameter models), but since it shows only a small number of tasks, it's only suggestive regarding in-context learning in general.

Recall that we define induction heads as heads that empirically copy arbitrary token sequences using a "prefix matching" attention pattern. Our goal is to find heads that meet this definition but *also* perform more interesting and sophisticated behaviors, essentially showing that induction heads in large models can be "generalizable".

In this argument, we show some anecdotal examples of induction heads from larger transformers (our 40-layer model with 13 billion parameters) that exhibit exactly such behaviors – namely literal copying, translation, and a specific type of abstract pattern matching. The behaviors are all of the form `[A*][B*]...[A][B]`, aka the "fuzzy nearest neighbor match" or "find something similar early in the sequence and complete the sequence in analogy". We verify that these heads score highly on our "copying" and "prefix matching" evaluations (that is, they increase the probability of the token they attend to, and attend to tokens where the prefix matches the present token on random text), and are thus "induction heads" by our strict empirical definition, at the same time as they *also* perform these more sophisticated tasks.

The results of some example heads are shown in the table below, and described in the subsections below.

Head	Layer Depth	Copying score (?)	Prefix matching score (?)
Literal copying head	21 / 40	0.89	0.75
Translation head	7 / 40	0.20	0.85
Pattern-matching head	8 / 40	0.69	0.94

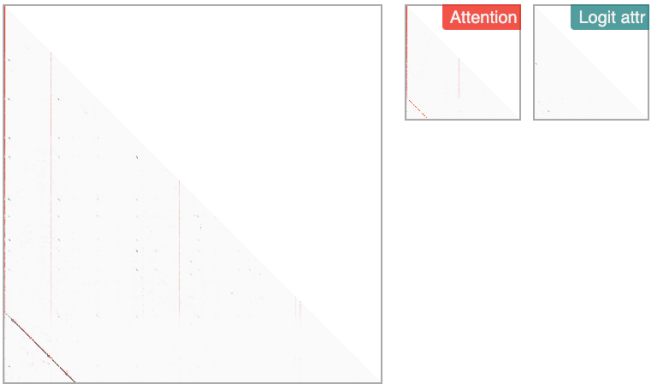
Behavior 1: Literal sequence copying

We'll start with the simplest case of a head that literally copies repeated text, to get familiar with the visualization interface we're using and the basic dynamics of these heads. We've selected an induction head which seems to perform very basic copying behavior and will look at how it behaves on the first paragraph of Harry Potter. We've repeated the first few sentences afterwards to show the head's behavior on longer segments of repeated text. ***For the following interactive visualizations, we recommend visiting the HTML article.***

The visualization will show two different things:

- In red, "Attention" lets you see where the head is attending to predict the *next* token.
- In blue, "Logit attr" shows the earlier tokens that contributed to the prediction of the *current* token, using "direct-path" logit attribution.[21]

To start exploring the visualization, we suggest you visit the HTML article and try hovering your cursor over the second paragraph.



If you explore the visualization, you'll see that the head predicts repeating the names "Dursley" and "Potters"; the phrase "a small son"; and then the entire repeated sentences at the end. In all cases, these successful predictions are made by attending back to a previous instance where this phrase was present in the text.

Behavior 2: Translation

It's a well-known result that language models can translate between languages. Intriguingly, we've encountered many examples of induction heads that can do translation. Here, we explore a head we found in layer 7 of our 40-layer model, showcasing translation between English, French, and German. (As with the others in this section, this head is *also* an “induction head” by the same definition we’ve been using all along, because when shown repeated *random* tokens, it uses a “prefix matching” attention pattern to copy the sequence verbatim.)